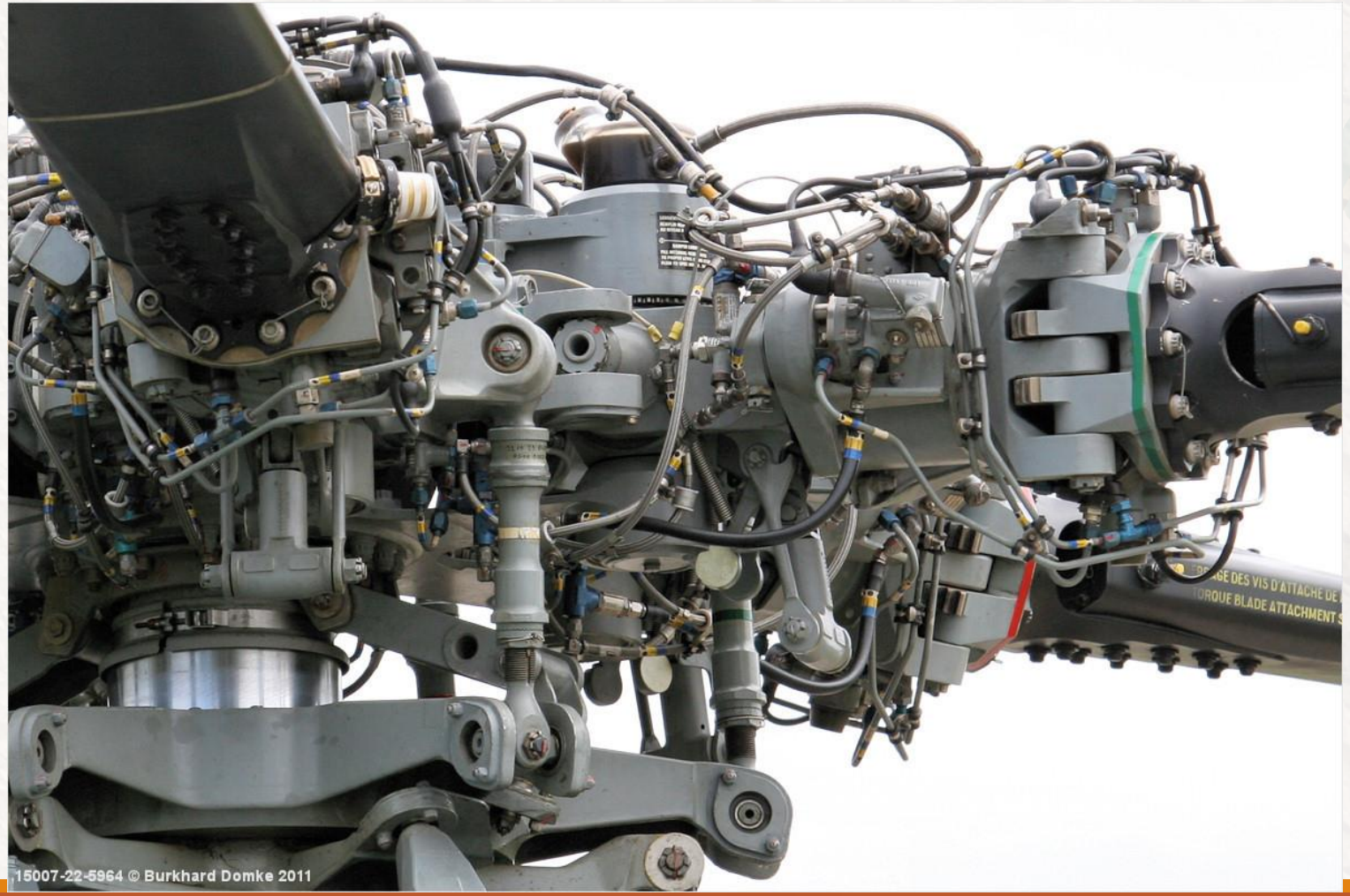
Rotary Wing Aerodynamics (AE 667)

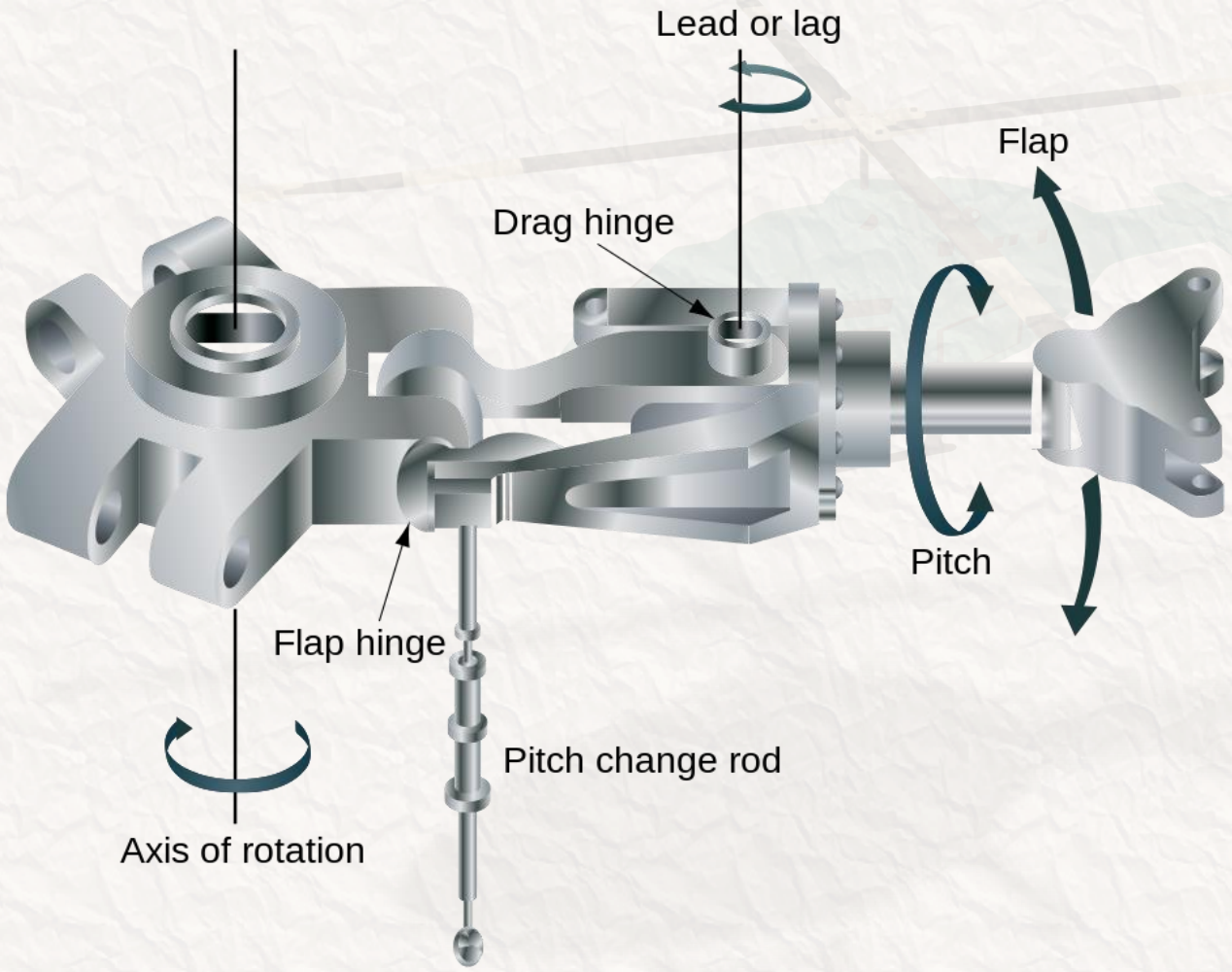


Blade Dynamics

Helicopter Rotor Hub



Helicopter Rotor Hub

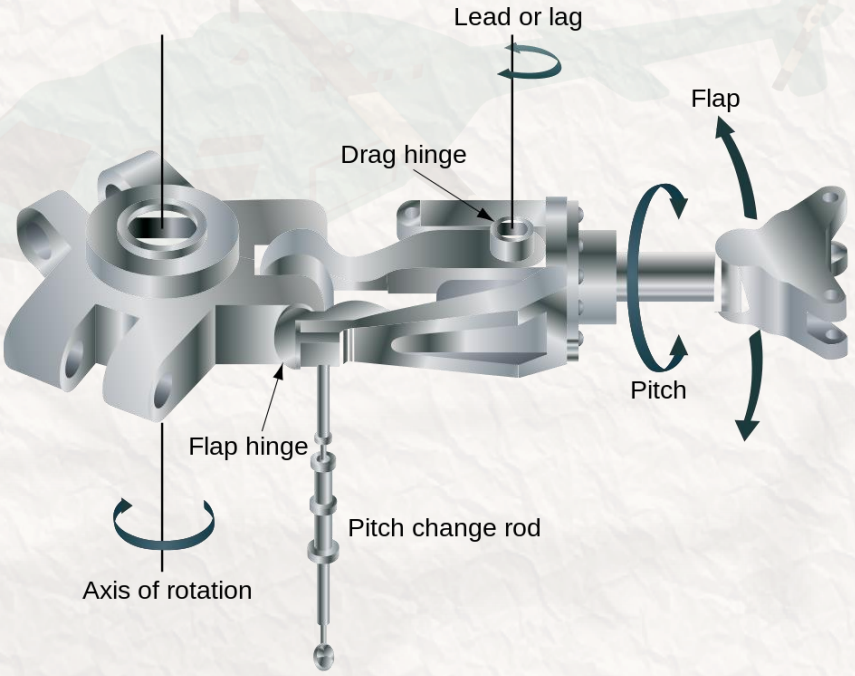


Helicopter Rotor Hub degrees of freedom

- Flapping hinges: Allows blade to flap up or down. Avoids transmission of bending moments to the shaft

- Lead-Lag hinges: Avoids stresses at the blade root due to Coriolis forces.
  (Lead-lag motion neglected from aerodynamic perspective)

- Pitch / Feathering (bearing, swash plate, pitch links): Control blade pitch for controlling helicopter motion.



Hub type: Rigid

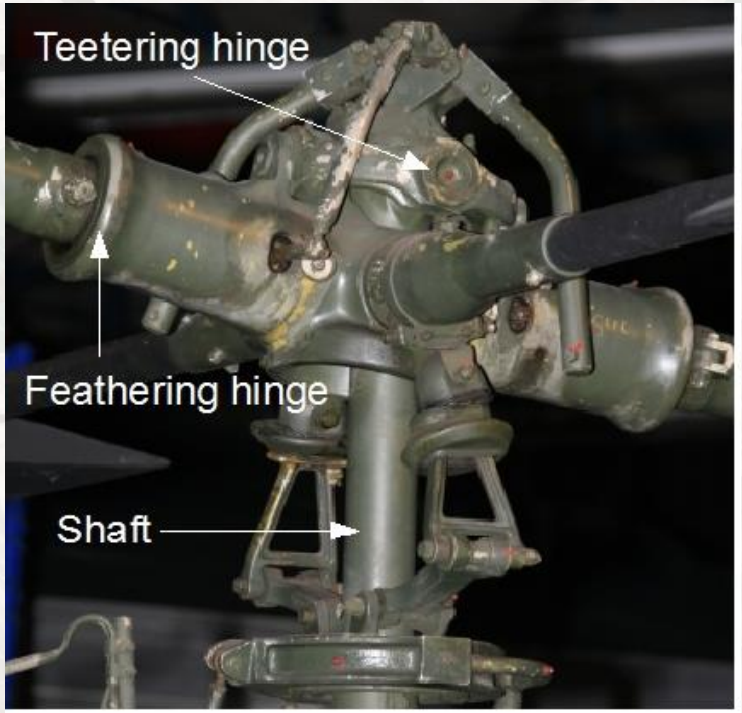
- Only feathering (pitching), No Flapping
- Moments transmitted to the rotor head
- High degree of control power.
- Fewer parts ➔ less maintenance



http://www.helistart.com/RotorheadTypes.aspx

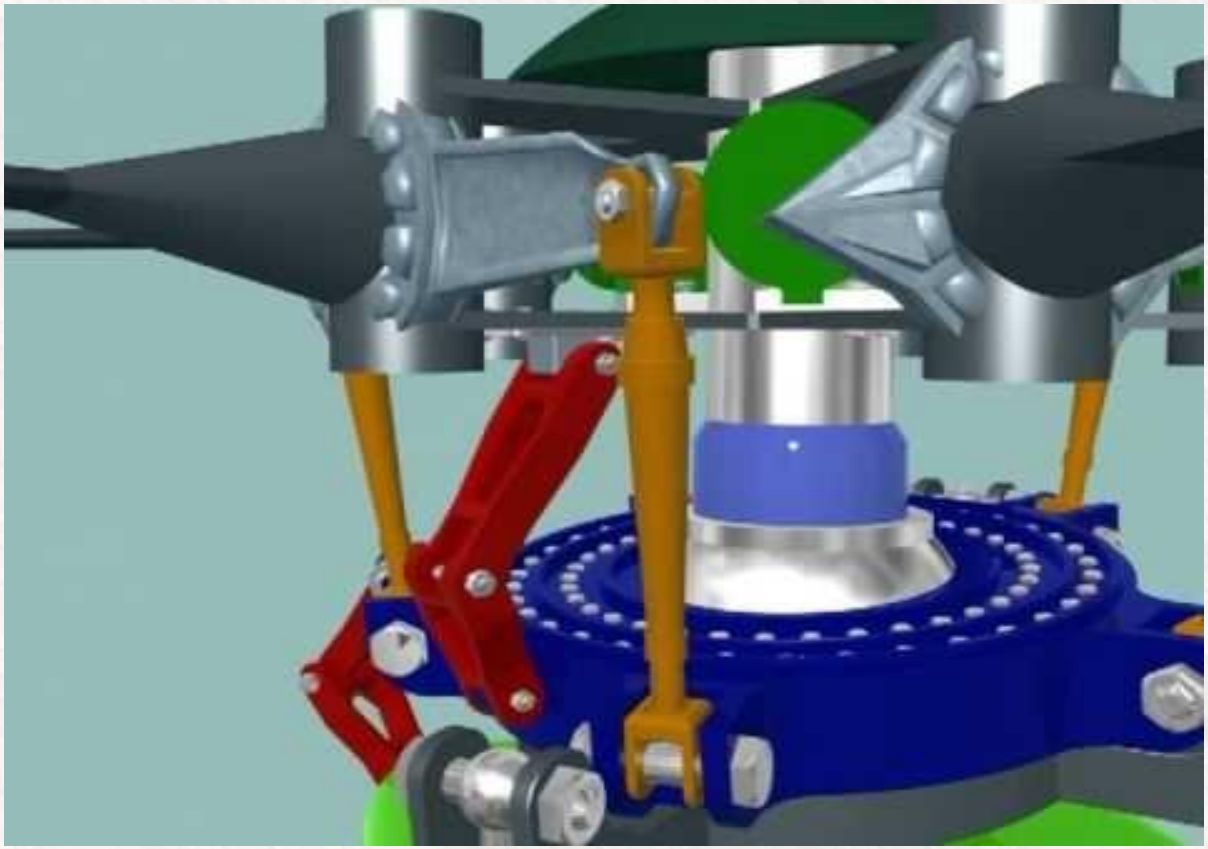
Hub type: Teetering

- Blades connected together through a common pivot
- Blades fixed at an average coning angle to reduce stress
- Rotor disc can tilt in any direction
- Lesser degree of control compared to rigid system



http://www.helistart.com/RotorheadTypes.aspx

Swash Plate Animation

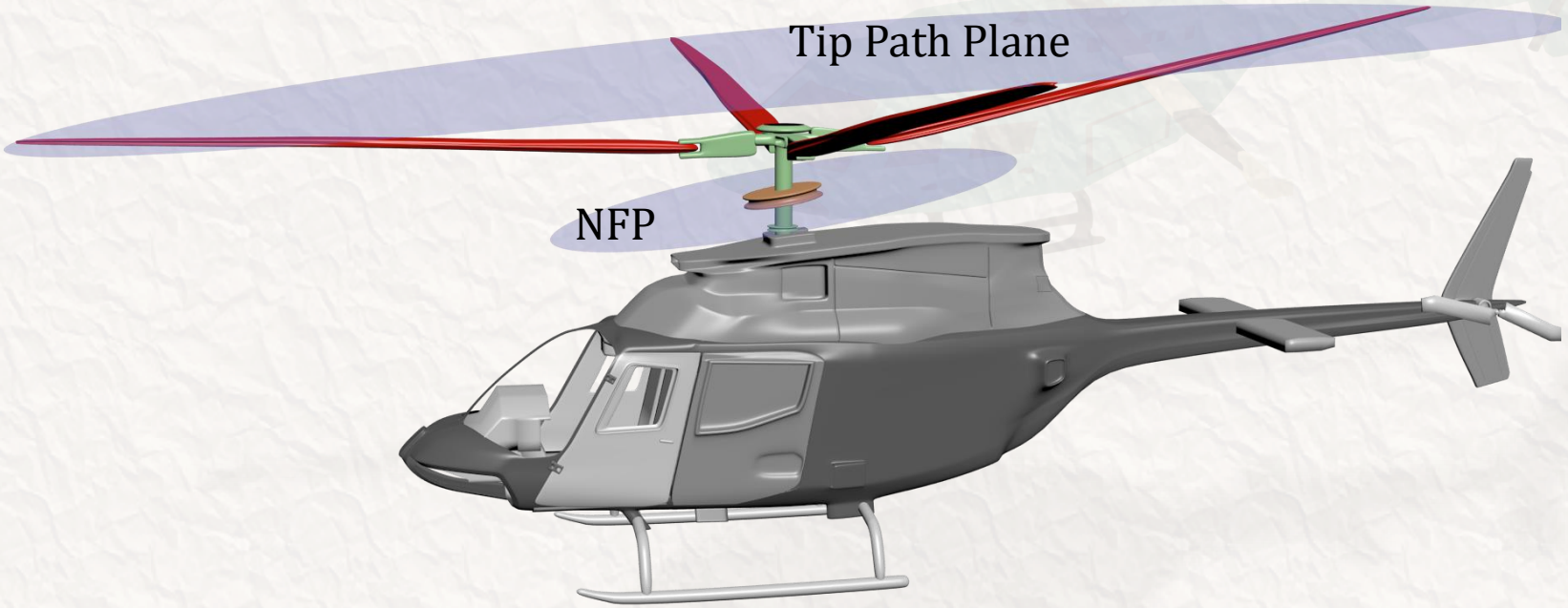


https://www.youtube.com/watch?v=83h6QK-oJ4M

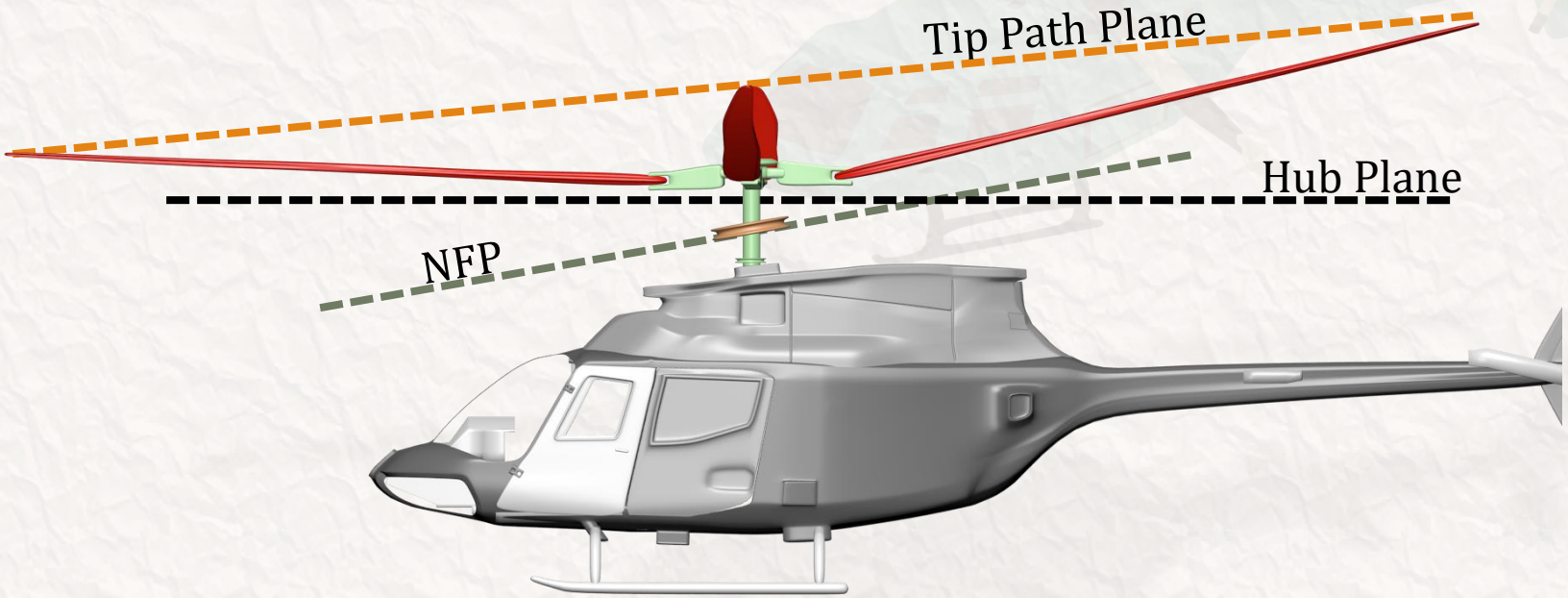
Coordinate Systems

- Hub Plane (Shaft Plane)
  - ‘Z’ axis along rotor shaft
  - Rotor is seen flapping and feathering (changes pitch) about this plane
- Tip Path Plane (TPP)
  - Plane in which the tips of the rotor blades move
  - Rotor is seen to be only feathering. No flapping motion
- No Feathering Plane (NFP)
  - Pitch constant when seen from this plane
  - Rotor is seen to be flapping only

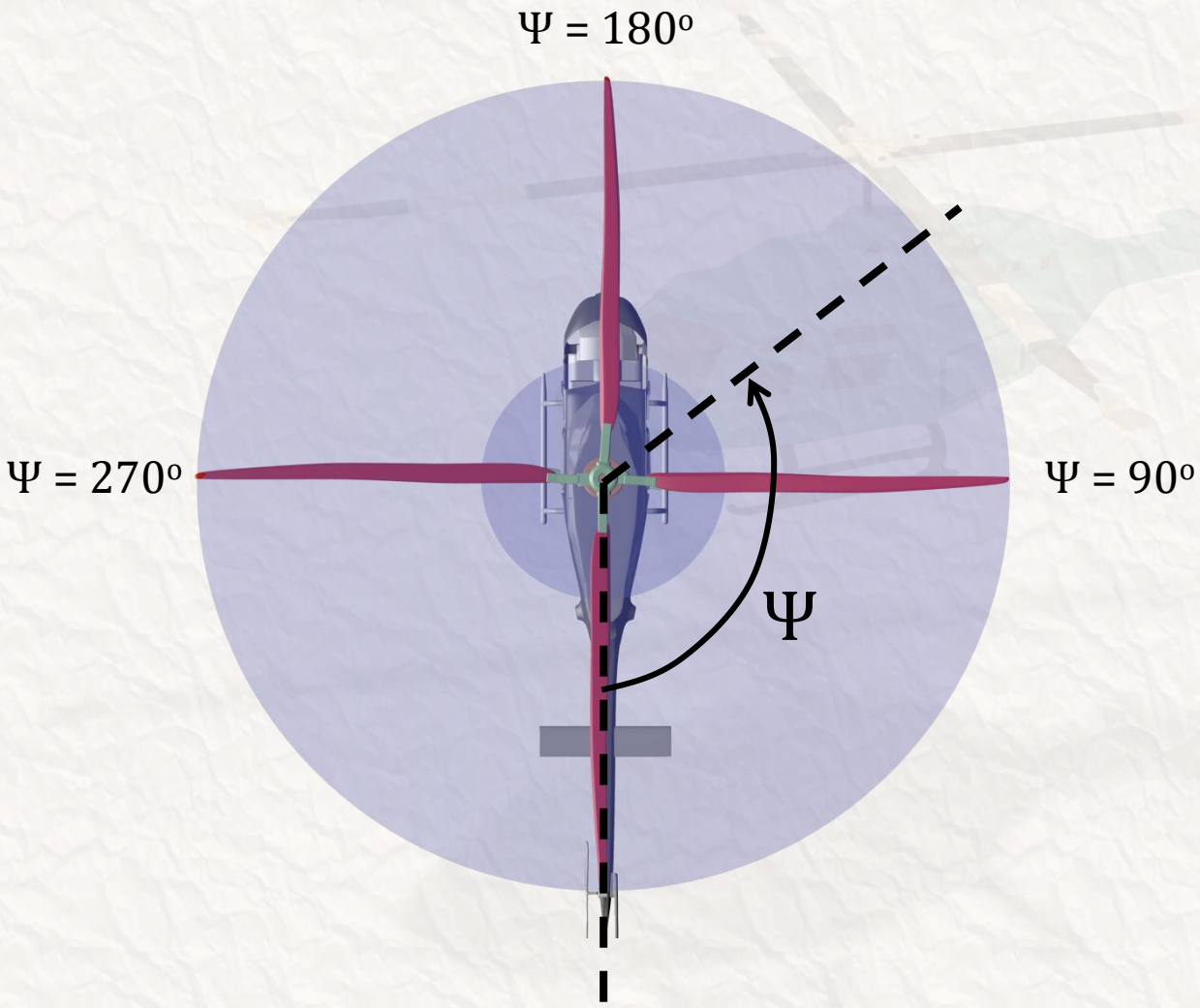
Coordinate Systems



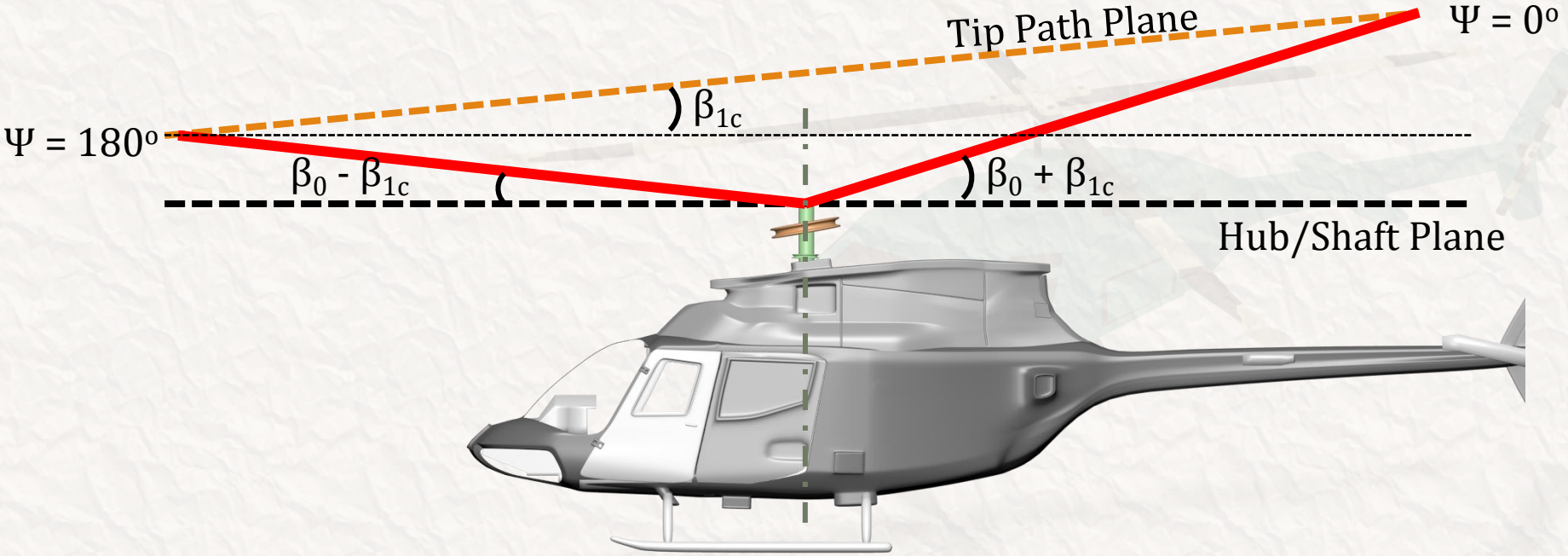
Coordinate Systems



Azimuthal angles



Tip Path Plane &Flapping Angle



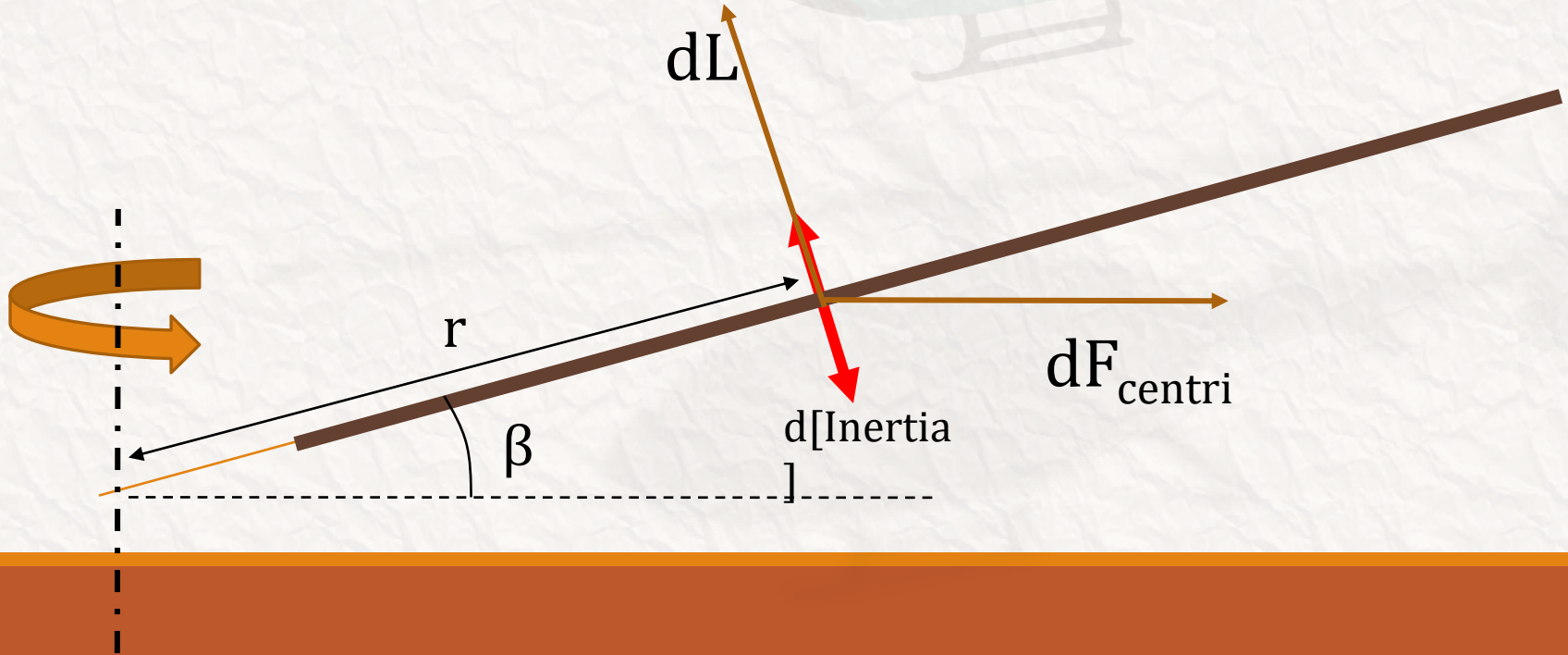
$$\beta\,(\text{Flapping Angle}) \cong \beta_0 + \beta_{1c}\cos(\psi) + \beta_{1s}\sin(\psi)$$

Higher order terms/harmonics relevant while studying vibration

Blade Dynamics
(Blade Flapping Dynamics)

Forces and Moments on Rotor Blade

- We assume 0 hinge offset for simplicity
- Steady forward flight: Net moments at hinge due to aerodynamic, centrifugal, & inertial forces cancel out to achieve equilibrium



Moment Balance

- Counter-clockwise moments
  - Aerodynamic:

$$\int_{r=0}^{r=R} \frac{1}{2} \rho c \left(\Omega r + V_\infty \sin\psi\right)^2 r C_l \, dr$$

(as derived while studying hover coning)

- Clockwise moments:
  - Centrifugal: $I\Omega^2\beta$ (as derived while studying hover coning)
  - Inertial
    (due to angular acceleration in flapping direction): $I\ddot{\beta}$

Moment Balance

- Clockwise Moments = Counter-clockwise moments

$$I\ddot{\beta} + I\Omega^2\beta = \int_{r=0}^{r=R} \frac{1}{2}\rho c\left(\Omega r + V_\infty \sin\psi\right)^2 r C_l dr$$

- The equation resembles a harmonic oscillator with natural frequency of Ω and the forcing term (right hand side) also consisting of first, second, and higher harmonics of Ω

- Aerodynamic damping ensures stability of the otherwise resonant system

Solving ODE (Partially)

- Assume forcing function (RHS) of the form: Asin(Ωt)
- Assume a damping term in LHS of form: c dβ/dt
- Resulting expression: $I\ddot{\beta} + c\dot{\beta} + I\Omega^2\beta = A\sin\Omega t$
- Assuming the solution to be of form:

$$\beta = B\sin\Omega t + C\cos\Omega t$$

- Substituting and solving, we get

$$\beta = -\frac{A}{c\Omega}\cos\Omega t = \frac{A}{c\Omega}\sin\left(\Omega t - \pi/2\right)$$

Solving ODE (Partially)

- Solution $\beta = \frac{A}{c\Omega}\sin(\Omega t - {}^{\pi}/_{2})$ for forcing field $A\sin(\Omega t)$

➔response lags 90° from excitation force.

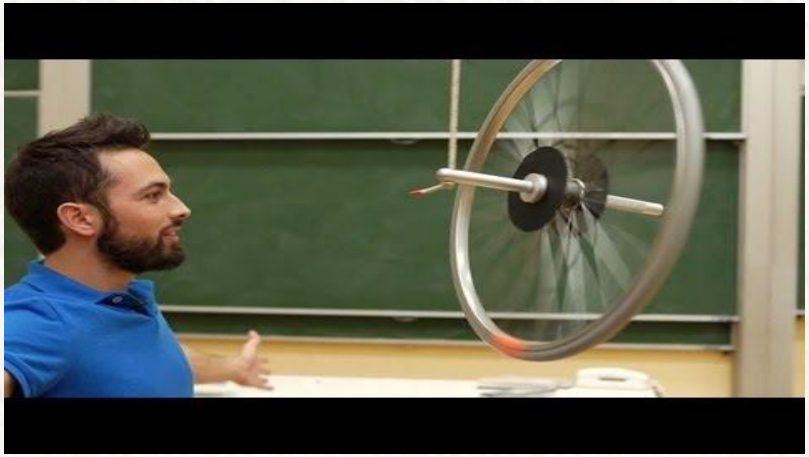
➔ the blades will flap upward to the highest level 90° after the highest pitch azimuth.

[Gyroscopic Precession!]

For example: In forward flight, the highest pitch setting is at $\Psi = 270^{o}$ and the lowest pitch setting is at $\Psi = 90^{o}$. Hence blades flap up at $\Psi = 0^{o}$ and down at $\Psi = 180^{o}$ (or in other words, TPP tilts forward). The overall thrust points forwards (to counter drag and to accelerate)

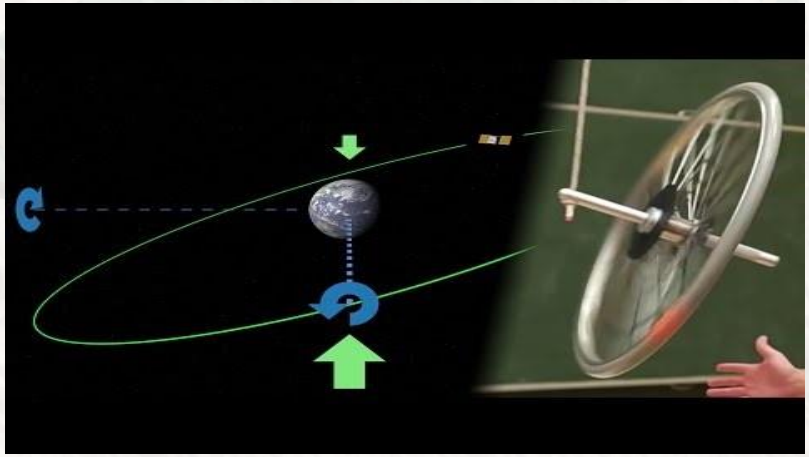
Gyroscopic Precession

- Please check out the following two videos on Gyroscopic Precession for a better understanding



Gyroscopic Precession

https://youtu.be/ty9QSiVC2g0



Gyroscopic precession -- An intuitive explanation

https://youtu.be/n5bKzBZ7XuM?t=204

General form of blade flapping motion

- The blade flapping dynamics ODE

$$I\ddot{\beta} + I\Omega^2\beta = \int_{r=0}^{r=R} \frac{1}{2}\rho c(\Omega r + V_\infty \sin\psi)^2 r C_l dr$$

- has solution of the form:

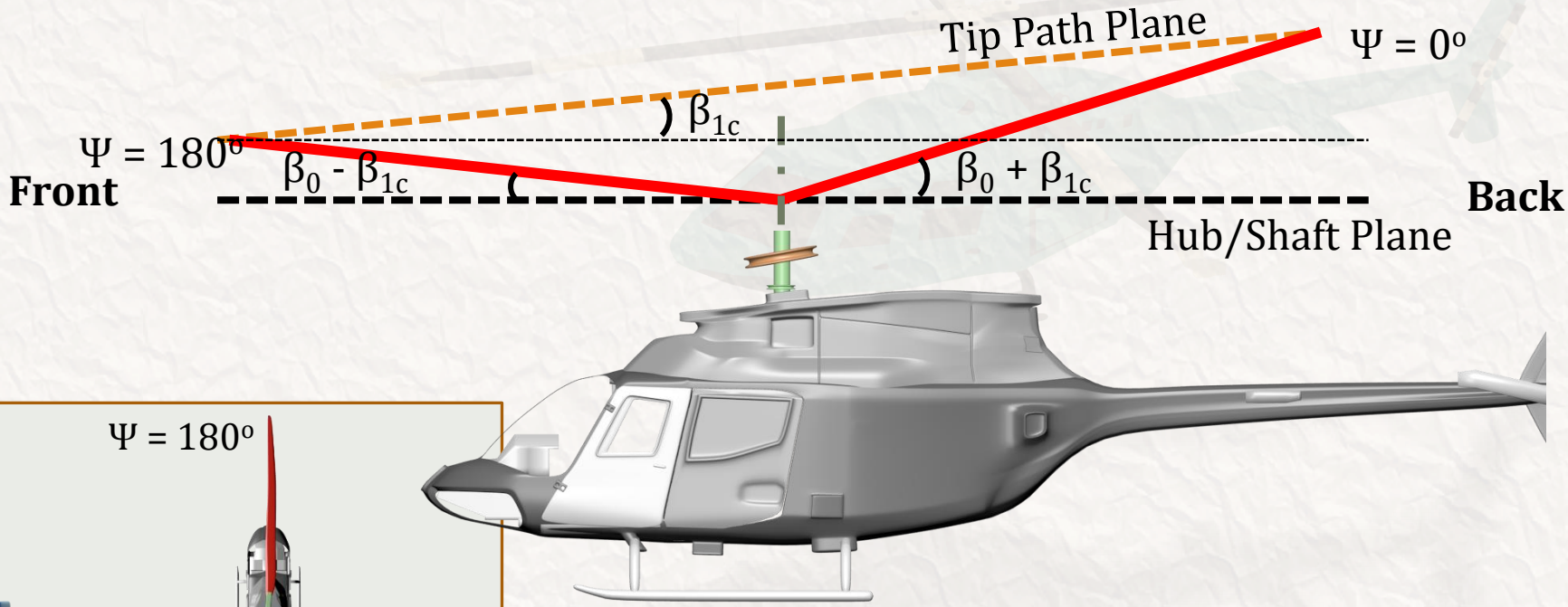
$$\beta = \beta_0 + \beta_{1c}\cos\psi + \beta_{1s}\sin\psi + \beta_{2c}\cos 2\psi + \beta_{2s}\sin 2\psi + \bullet\bullet\bullet$$

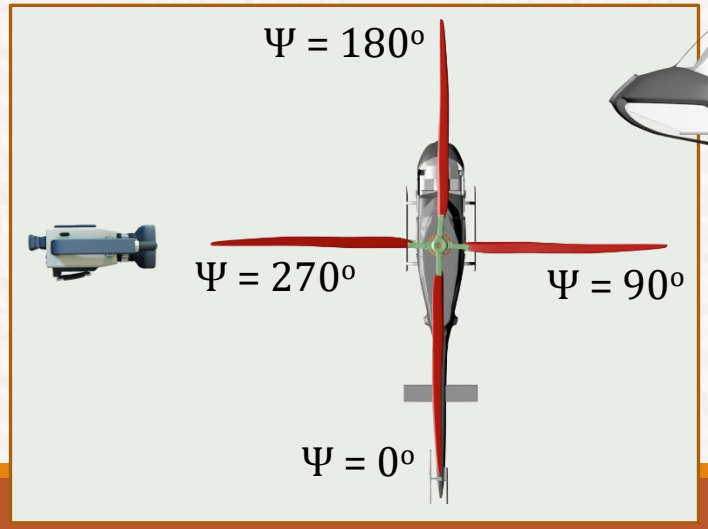
- For aerodynamics purpose, terms upto the first harmonic ψ are sufficient. Higher harmonics (2ψ and above) are important from dynamics and vibration perspective

Note that β_0 is the coning angle and $\psi = \Omega t$

Significance of β_{1c}

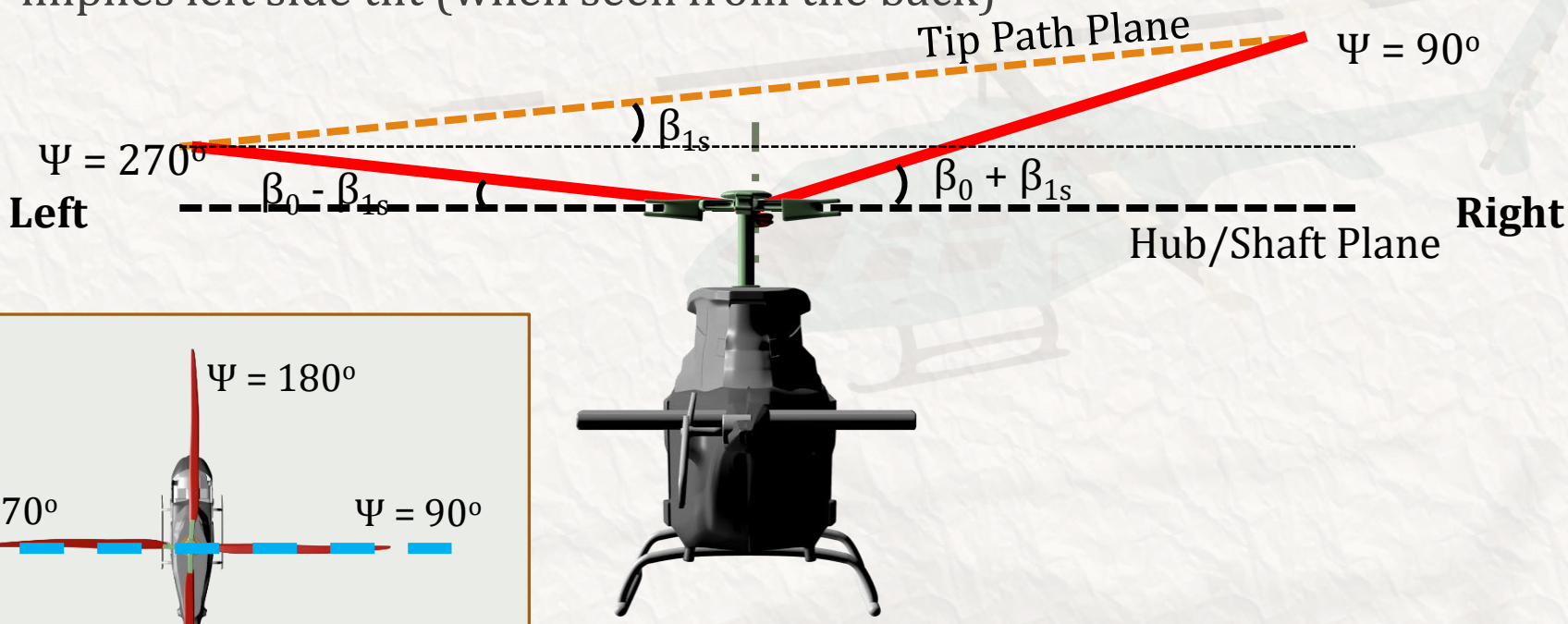
- **β_{1c}** is how much the TPP tilts forwards or backward w.r.t shaft plane. Positive β_{1c} implies forward tilt

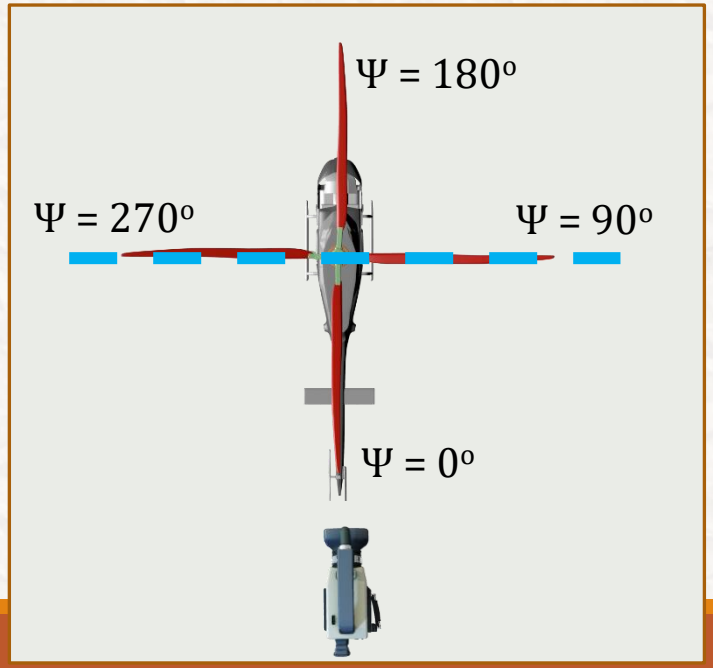




Significance of β_{1s}

- **β_{1s}** is how much the TPP tilts sidewards w.r.t shaft plane. Positive β_{1s} implies left side tilt (when seen from the back)





Pitch Control (Pilot Input)

- Pilot controls collective and cyclic pitches of the blades through swashplate.

$$\theta_{75\%R} = \theta_0 + \theta_{1c}\cos\psi + \theta_{1s}\sin\psi$$

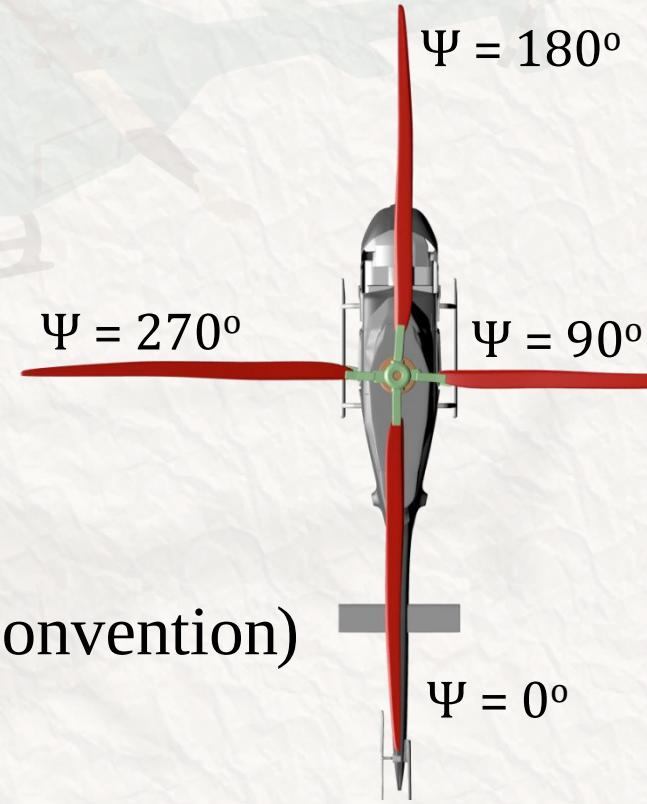
Where,

θ_0 = Collective Pitch

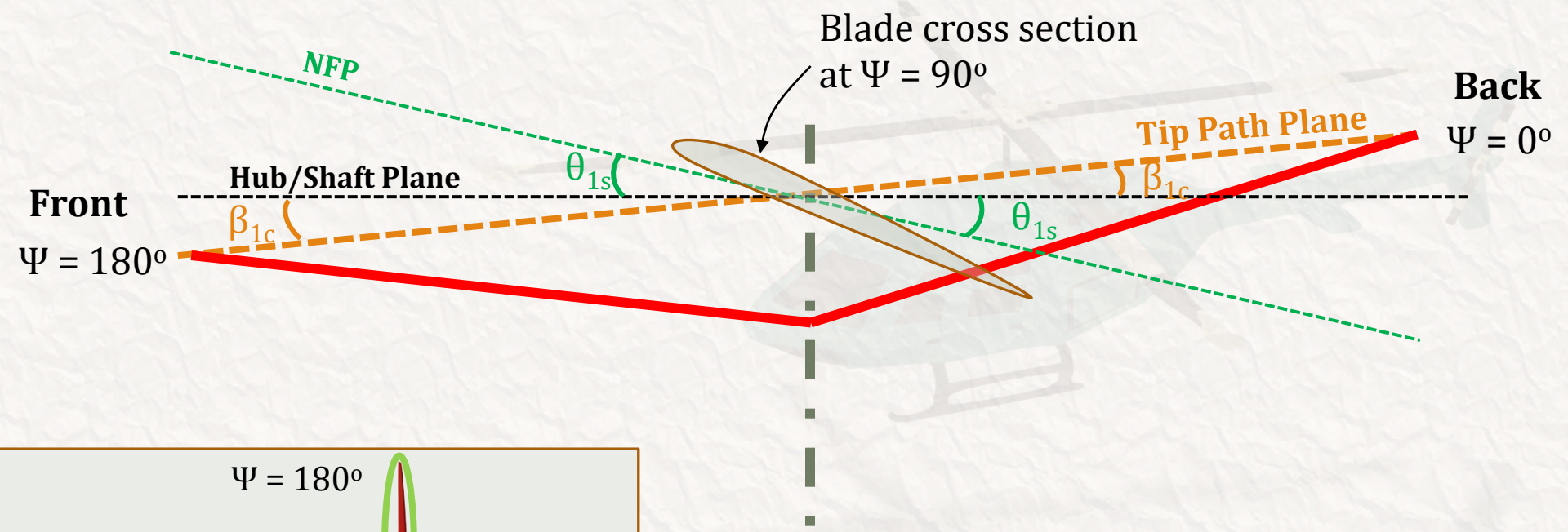
θ_{1c} = Lateral Cyclic (Side wards)

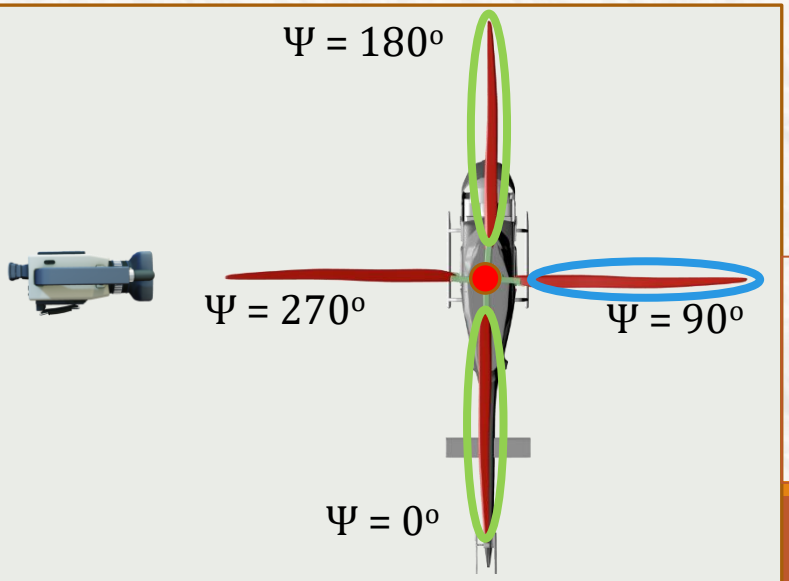
θ_{1s} = Longitudinal Cyclic (Fore & Aft)

$\theta_{75\%R}$ = Pitch at 75%R (definition as per convention)



Longitudinal Control and Angles

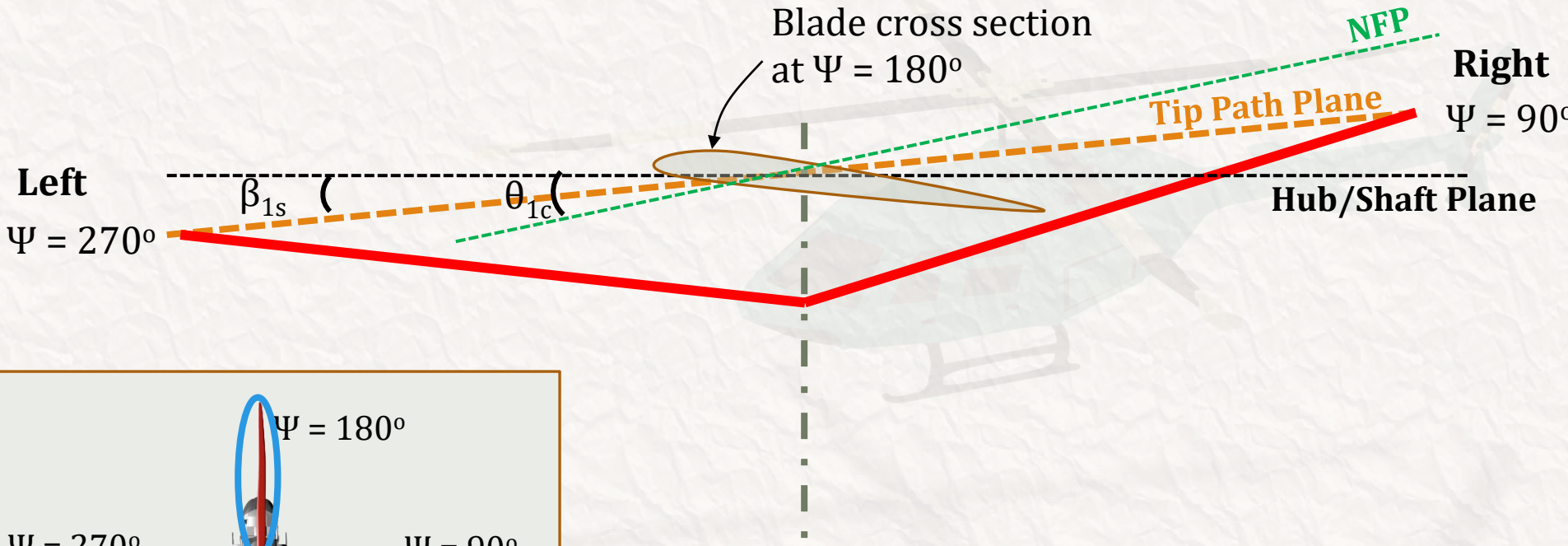


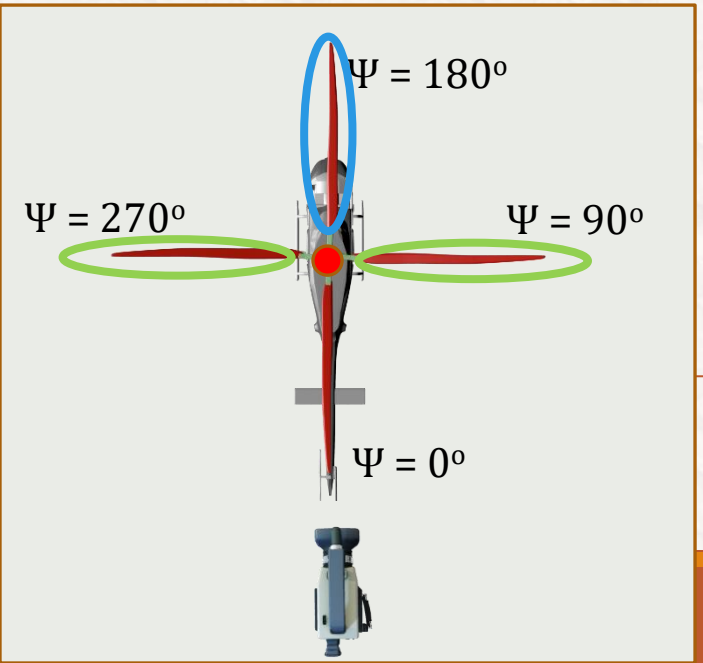


$$\theta = \theta_0 + \theta_{1c}\cos\psi + \theta_{1s}\sin\psi$$
$$\theta_{\psi=90^o} = \theta_0 + \theta_{1s}$$

$\theta_{1s} + \beta_{1c}$ is independent of coordinate system of measurement
(as the relative angle between the NFP and TPP does not change with coordinate system)

Lateral Control and Angles





$$\theta = \theta_0 + \theta_{1c}\cos\psi + \theta_{1s}\sin\psi$$

$$\theta_{\psi=180^o} = \theta_0 - \theta_{1c}$$

θ_{1c} - β_{1s} is independent of coordinate system of measurement (as the relative angle between the NFP and TPP does not change with coordinate system)

Angles in different coordinate systems

- $\theta|_{TPP} = \theta$ pitch as observed from Tip Path Plane
- $\beta|_{NFP} = \beta$ flap angle as obsered from No Feathering Plane

- $\beta_{1c} + \theta_{1s} = \beta_{1c}|_{NFP} = \theta_{1s}|_{TPP}$
- $\beta_{1s} - \theta_{1c} = \beta_{1s}|_{NFP} = -\theta_{1c}|_{TPP}$

- $\beta|_{TPP} = \beta_0$ and $\theta|_{NFP} = \theta_0$ by definition

$(\beta_{1c} + \theta_{1s})$ and $(\beta_{1s} - \theta_{1c})$ will always appear as single terms in the future derivations. That is so, because aerodynamics on a blade section is independent of whether the resultant pitch is due to direct pilot input (θ)or due to tilting of the TPP (β)